Branch Closures and Evictions in the US

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Abstract: More than two million evictions occur in the US each year, a phenomenon with profound and lasting consequences for the families and communities affected. This paper examines how the closure of brick-and-mortar bank branches in the US has affected the number of households threatened with evictions in the period 2000-2018. To overcome the potential endogeneity associated to branch closures we adopt an instrumental variable (IV) identification strategy that uses bank mergers as an instrument of the distance between the population and bank branches at the census tract level. Our results show that both interstate and intercounty mergers had a positive and significant effect in the distance to the closest bank branches. Moreover, we find that a 1% increase in the distance to the closest branch generated a 2.3% increase in the number of households threatened with an eviction. This result is primarily driven by the effects in urban tracts. We complement our analysis with several robustness checks, including a matching procedure to control for pre-existing observable differences between tracts exposed and unexposed to mergers. Moreover, we re-estimate our model focusing on mergers where the acquiring and the acquired banks had overlapping branches within the same tract, and we consider different measures of financial exclusion and of exposition to evictions.

Keywords: Evictions, Financial Exclusion, Bank branches, Mergers, United States
JEL Codes: L16; L22; G21; G34; G38

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1. Introduction

Eviction is one of the most relevant and prevalent social problems in the United States. More than two million of families across the country are evicted every year.[1] An eviction occurs when the owner of a property expels tenants from their residence through a formal judicial procedure. The primary driver for evictions is that tenants do not (cannot) pay their rent. This usually affects low-income families that qualify for affordable housing programs, but do not get any help (Desmond, 2012; Desmond et al., 2015; Desmond and Gershenson, 2016; Desmond, 2016; Desmond and Gershenson, 2017; Collinson et al. 2022). Poor women of color, domestic violence victims and families with children are also at high risk for eviction (Desmond, 2012; Desmond et al., 2013; Desmond, 2016).

Evictions can have deep and long-lasting consequences for the households affected. Recent academic research has highlighted both its material and health consequences. Evicted families often lose their possessions, are displaced from their communities, and see that their children are forced to change school (Been et al., 2011). Evictions have been associated with increased exposition to poverty (Desmond, 2012 and 2016; Desmond and Shollenberger, 2015; Collinson et al. 2022), unemployment (Folwer et al. 2015; Desmond and Gershenson, 2016; Collinson et al. 2022) and crime (Falcone, 2022).[2] Furthermore, mothers who have experienced eviction report worsened health for both themselves and their children, as well as more parenting stress (Desmond and Kimbro, 2015; Currie and Tekin, 2015; Ramphal et al. 2023).

Economic research has explored various factors contributing to this social problem (Crane and Warnes, 2000; Desmond and Kimbro, 2015; Collinson et al., 2022). However, to the best of our knowledge, no studies have investigated the impact of financial exclusion and changes in the bank market on the prevalence of eviction lawsuits and the disparities in eviction filings across local communities. The purpose of our paper is to examine whether the consolidation of the bank market and the reduction in the number of bank branches in the US has

[1] According to Gromis et al. (2022), in an average year between 2000 and 2018, landlords filed more than 3.6 million eviction cases, resulting in almost 7% of renting households facing an eviction lawsuit.

[2] Desmond (2012) explains that "*because many landlords reject applicants with recent evictions, evicted tenants are pushed to the very bottom of the rental market and often are forced to move into run-down properties in dangerous neighborhoods*".

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contributed to an increase of evictions. Specifically, we analyze whether the closure of branches and the increase of financial exclusion following the 2007-2008 financial crisis led to an increase of evictions at the census tract level.

Over the past few decades, the United States banking market has undergone significant consolidation, resulting in fewer banking entities and a transformation of their branch networks. The Riegle-Neal Act of 1994 introduced market liberalization, leading to an increase in the number of bank branches as banking entities expanded into other states. However, this trend reversed after the 2007-2008 financial crisis and the Great Recession. From 2009 to 2020, the US saw a loss of fifteen percent of bank branches (Calzada et al., 2023). During this period, numerous urban areas experienced high levels of branch closures while many rural areas became "banking deserts". This trend is expected to continue in the coming years as the banking sector embraces digitalization and FinTech services.

In recent years, mobile and internet banking have been replacing branch-based services. However, concerns remain that branch closures may heighten the risk of consumers becoming unbanked or underbanked (Googstein & Rhine, 2017; Célerier & Matray, 2019). The absence of a nearby bank branch can raise consumer costs associated with accessing financial services, such as cashing checks or applying for credit. This "financial exclusion" can be particularly significant for low-income families and minority groups, who may have limited mobility and internet access (Sinclair, 2001; Carbo et al., 2007; Degryse & Ongena, 2005; Ergungor, 2010; Nguyen, 2014).

Additionally, limited access to bank branches may drive consumers to depend on alternative, unregulated financial providers, such as check-cashing outlets, payday lenders, or auto loan title lenders, which can impose exorbitant or predatory fees (Caskey, 1994 & 2002; Bass & Campbell, 2013; Edlebi et al., 2022). Another notable consequence of branch closure is the potential erosion of personal relationships between individual borrowers and loan managers, which are crucial for credit access. Branch closures often result in staff reallocation, making it difficult for consumers to establish new personal relationships with personnel at other branches or banks (Drexler & Schoar, 2014; Nguyen, 2019). Furthermore, obtaining financial advice about products and services may be more challenging through online banking (Federal

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Reserve Board, 2016).[3] Considering this, the aim of our study is to examine whether poorer access to bank branches increases the likelihood of eviction for the affected population.

Our paper examines how the closure of brick-and-mortar bank branches in the United States before and after the 2007 economic crisis has affected the number of households threatened with evictions. One obvious difficulty in estimating the effects of branch closures is that the decision to close a branch may be correlated with unobserved characteristics of the local market and with the socio-economic characteristics of the neighborhood. For example, a rise in unemployment within a specific area may result in banks choosing to close a branch, as well as a surge in eviction filings due to unpaid rent. To overcome this endogeneity problem, our paper adopts an instrumental variable (IV) identification strategy that uses bank mergers as an instrument for the distance between the population and bank branches at the census tract level. Following Nguyen (2019), our estimation strategy exploits within-tract differences in exposure to mergers of *large banks*. We consider that the mergers of *large banks* are plausibly exogenous to local economic conditions and we use them as instruments for branch closure at the census tracts level (Nguyen, 2019; Jagtiani and Maingi, 2019; Calzada et al. 2023). In addition, to guarantee that mergers are unrelated to unobserved local market characteristics, we restrict our analysis to 106 mergers involving entities with more than $1 billion in pre-merger assets and where the acquired bank operated more than 80 branches. Moreover, we focus on mergers between banks in different states (interstate mergers) and between banks in the same state but with their headquarters in different counties (intrastate/inter-county mergers).

Our analysis draws on a rich dataset for 72,738 census tracts located in the 50 states of the United States and the District of Columbia. We use information on bank branches and mergers from the Summary of Deposit (SOD) and the Reports of Structure Changes (ROC) provided by the Federal Deposit Insurance Corporation (FDIC). Data on evictions has been obtained from the Evictions Lab at Princeton University (Desmond et al. 2018; Gromis et al.

[3] Googstein and Rhine (2017) show that while the distance to bank branches can help explain why some households do not have a bank account or use alternative financial services in addition to having a bank account, this aspect is relatively un-important compared to other demand- and supply-side factors.

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2022a and 2022b). Tract-level demographic and socio-economic variables are taken from the US Census Bureau.

Our results of the first stage of the IV estimation shows that both interstate and intercounty mergers had a positive and significant effect in the *distance* of the population to the closest bank branch.[4] These results suggest that in the period considered mergers had a negative effect on financial exclusion, as they increased the distance that consumers had to travel to receive bank services. Moreover, the result of the second stage of the estimation shows that the distance to the bank branches had a positive and significant impact in the number of households threated with evictions. Specifically, we find that a 1% increase in the distance to the closest branch generated a 2.3% increase in the number of households filled with an eviction. This result is mostly due to the effects of financial exclusion on urban tracts. Indeed, in the case of urban tracts a 1% increase in the distance increased the number of households threatened with an eviction in 3%. In contrast, we do not find a statistically significant relationship between distance and eviction for rural tracts.

We repeat this analysis dividing the sample in three sub-periods, 2000-2007 and 2008-2013 and 2014-2018. The first interval corresponds to a period of expansion of bank branches and the other two to a period of branch consolidation after the financial crisis of 2008. Our results show that the distance to bank branches had a positive impact on evictions in the sub-periods 2000-2007 and 2008-2014. Specifically, a 1% increase in the distance to the closest branch generated a 5% and 1.7%, respectively, increase in the number of households threatened with an eviction. This implies that in the period of branch expansion (2000-2007) the presence of more branches could increase the financial support received by tenants and this could reduce the likelihood of evictions. Likewise, in the period after the financial crisis (2018-2013) branch closing could exacerbate tenants' financial difficulties and make them more exposed to evictions. Finally, we do not find a significant relationship between distance and evictions

[4] Our paper focusses on the distance between bank branches and the potential borrowers (operational distance). By contrast, Zhao et al. (2021) focus on the impact of functional distance (distance between bank branches and bank headquarters) on soft information. Using data on lending by British banks, they show that functional distance is negatively related to the change in SME lending, but has no impact on the change in mortgage lending. Bernini and Brighi (2018) examine Italian cooperative banks from 2006 to 2013 to investigate the effects of branch network expansion (which increased functional distance) on banks' efficiency costs.

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in the period 2014-18, although this result should be interpreted with caution, as mergers in this period are not a good instrument for branch closures.

We completed our analysis with several robustness checks. First, we apply a matching procedure to control for the existence of pre-existing observable differences between tracts exposed and unexposed to mergers that might impact our results. The adoption of the matching process allows us to select a more comparable group of tracts exposed and unexposed to mergers. Specifically, we use a propensity score matching algorithm to individually assign to the tracts affected by mergers other comparable tracts unaffected by mergers. Second, we re-estimate our baseline model focusing on mergers where prior to the transaction the acquirer and the acquired banks had overlapping branches within the same tract. Indeed, we expect that in tracts with overlapping bank branches, the merged entity will be more likely to close branches. Third, we use as an alternative measure of financial exclusion, the number of branches per inhabitant at the census tract level (instead than the distance to the closest bank branch). Finally, we re-estimate our model using alternative variables to measure the population exposed to evictions. Specifically, while the dependent variable in our baseline model is the *number of households threatened with eviction*, in these robustness checks we use the *total number of evictions* at the census tract-year level and the *number of eviction filings resulting in eviction judgments*. The results obtained from all these robustness regressions are very similar to those obtained in the baseline model.

Overall, our paper highlights the importance of financial exclusion for tenants. The availability and geographic proximity of bank branches shape the allocation of credit to tenants and their capacity meet rent obligations. In this context, branch closures have significant implications. When a branch closes, soft information about tenants is often not transferred to other branches or potential lenders. We believe that our findings can help public authorities at the federal, state and local level in designing policies aimed at preventing or reducing evictions.[5] Given the serious disruptions caused by evictions – such as homelessness, income loss, and health problems - public policies could be adopted to offer

[5] Lee and Bostic (2020) alert about the consequences of public policies aimed at promoting credit flows to lower-income and minority neighborhoods. They examine the lending behavior of U.S. ban, focusing on the unintended effects of the Community Reinvestment Act (CRA), a regulation that encouraged credit flows to historically disinvested neighborhoods. Kim (2023) examines the effects of the CRA on business employment.

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greater leniency in eviction proceedings (emergency support, legal assistance) and to mitigate the consequences of branch closures.

The article is structured as follows. Section 2 reviews the literature closely related to our paper. Section 3 describes the evolution of the US banking market in the last two decades. Section 4 explains the eviction process in the United States. Section 5 describes the dataset and section 6 explains our empirical strategy. Section 7 presents the results of our analysis, while section 8 shows several robustness checks. Finally, Section 8 concludes.

2. Literature review

Our paper contributes to the literature examining the drivers of evictions (Crane and Warnes, 2000; Desmond and Kimbro, 2015; Collinson et al., 2022).[6] Nonpayment of rent is a frequent cause of eviction, although not all tenants experiencing delays in their payments are evicted. Desmond and Gershenson (2016) analyze the factors associated with evictions in Milwaukee using discrete hazard models. They find that the number of children, race, job loss, and rental payment history are robust predictors of eviction, while renters in gentrifying neighborhoods do not have a higher eviction risk. Families in disadvantaged neighborhoods have a higher probability of eviction compared to those in transitioning areas. Moreover, tenants facing poverty shocks (e.g., eviction, incarceration, teenage pregnancy) or current disadvantages (e.g., joblessness, addiction, abusive relationships) are more likely to be evicted. Collinson et al. (2022) examine tenants' outcomes before and after their eviction cases, showing that those appearing in housing courts have lower earnings, employment, and credit access, along with higher levels of debt in collections. Both evicted and non-evicted tenants experience significant declines in earnings, employment, and credit scores, as well as increases in hospital visits, unpaid bills, and payday loan inquiries in the two years before the eviction case. Using an instrumental variables (IV) approach based on the random assignment of judges with varying eviction tendencies, they find that eviction orders lead to higher rates of homelessness, reduced earnings, lower durable consumption, and decreased access to credit.

[6] There is also a close literature examining the effect of foreclosure on households, which occurs when banks or other lending agencies initiate a process to repossess a home (Currie and Tekin, 2015; Molloy and Shan, 2013; Diamond et al., 2020).

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Other papers have examined the impact of regulatory changes on eviction rates. Gallagher et al. (2019) find that expansions of the Patient Protection and Affordable Care Act (ACA) lowered the cost of health insurance and reduced eviction-filing rates. Zewde et al. (2019) show that these regulations led to a reduction in home evictions and eviction filings per 1,000 renter-occupied households across U.S. states. Kroeger and La Mattina (2020) examine the causal impact of nuisance ordinances on evictions by exploiting the growth in nuisance ordinances in Ohio. Difference-indifferences estimates show that the municipalities that pass nuisance ordinances exhibit significant increases in both eviction filings and complete evictions. Using a similar approach, Falcone (2022) examines the impact of evictions on crime, using the staggered adoption of nuisance ordinances across Ohio cities from 2000 to 2014. The study finds that nuisance ordinances increased evictions by 32% and led to a rise in crime, including 18% more burglaries and 28% more vehicle thefts, as evictions lowered the private opportunity cost of committing crimes. Gromis et al. (2022) show that policies requiring landlords to provide a notice period before filing for eviction significantly reduce the number of eviction cases. In states with mandatory notice periods, tenants are less likely to have an eviction filing in their housing history. This suggests that tenants need time to collect the money to pay the rent, which is the time needed to talk to family, friends or financial institutions.

Our paper is also related to the literature on the branching strategies of US banking entities. Most studies in this area examine the impact of branch competition on prices (Sapienza, 2002), service quality (Focarelli and Panetta, 2003; Panetta et al., 2009), firm-bank relationships (Karceski et al., 2005; Panetta et al., 2009), consumer bankruptcies (Allen et al., 2016), bank efficiency and stability (Calomiris, 2000; Aubuchon and Wheelock, 2010; Aguirregabiria et al., 2016), and on the influence of market regulations (Adams and Amel, 2007; Cohen and Mazzeo, 2010; Calzada et al. 2023). Our paper is more closely aligned with the literature analyzing the effects of debranching on financial exclusion. Bank branches can help reduce information frictions by collecting soft information about consumers and their neighborhoods. Googstein and Rhine (2017), using 2011 FDIC and US Census Bureau data, show that the proximity to bank branches helps explain why some households are unbanked or use alternative financial services. However, they highlight that distance to the branches is less important compared to other factors like income, education or race. Célerier and Matray

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(2019) analyze the impact of branch expansion following the Riegle-Neal Interstate Banking and Branching Efficiency Act of 1994. Using FDIC data and the staggered implementation of the Act, they show that between 1993 and 2005, branch density grew by approximately 20%, and the probability of financial inclusion by low-income household increased by 4%. Other studies have demonstrated that the distance between lenders and borrowers plays a crucial role in determining credit availability and terms, particularly in low-income neighborhoods (Degryse and Ongena, 2005; Agarwal and Houswald, 2010; Ergungor, 2010; Brown, 2016; Allen et al., 2012; Beck et al., 2019; Martin-Oliver, 2019). [7]

Another area of research that is relevant for this paper is the one that analyzes the relationship between mergers and financial exclusion. Allen et al. (2015) examined the impact of the merger between Toronto Dominion and Canada Trust in 2000 on consumer bankruptcy. Beck and Martínez Peria (2010) explored how foreign bank acquisitions affected banking outreach in Mexico between 1997 and 2005. Jagtiani and Maingi (2019) investigated how community banks mergers affected small business lending in the period from 2002 to 2014. Finally, our paper closely follows Nguyen (2019), who estimated the causal impact of bank branch closings during the period 1999-2012 on local access to credit. Using exposure to merger induced consolidation as an instrument for branch closings, she shows that closings caused a sharp decline in credit supply to local small businesses but had no effect on mortgage lending. She concludes that "distance matters, not only because it improves accessibility, but also because it reduces the costs of transmitting information and facilitates the forging of long-term relationships".

3. Evictions

Eviction cases are civil lawsuits filed by landlords to remove tenants or collect unpaid rent or other monetary damages (Eviction Lab).[8] These proceedings are typically initiated for

[7] A recent literature examines the impact of urbanization on access to credit. Gabriel and Rosenthal (2013) find that urbanization increases home mortgage origination rates and loan amounts. Lee and Luca (2018) show that firms in large cities are less likely to experience credit constraints. Carmignani et al. (2021) find that urbanization positively affects credit access for construction firms but not for manufacturing or service firms.

[8] Foreclosures, in contrast, occur when a lending institution requires the owner or the occupant of a property to leave. According to Desmond and Kimbro (2015), while evictions tend to affect the urban poor, foreclosures usually affect the working and middle class.

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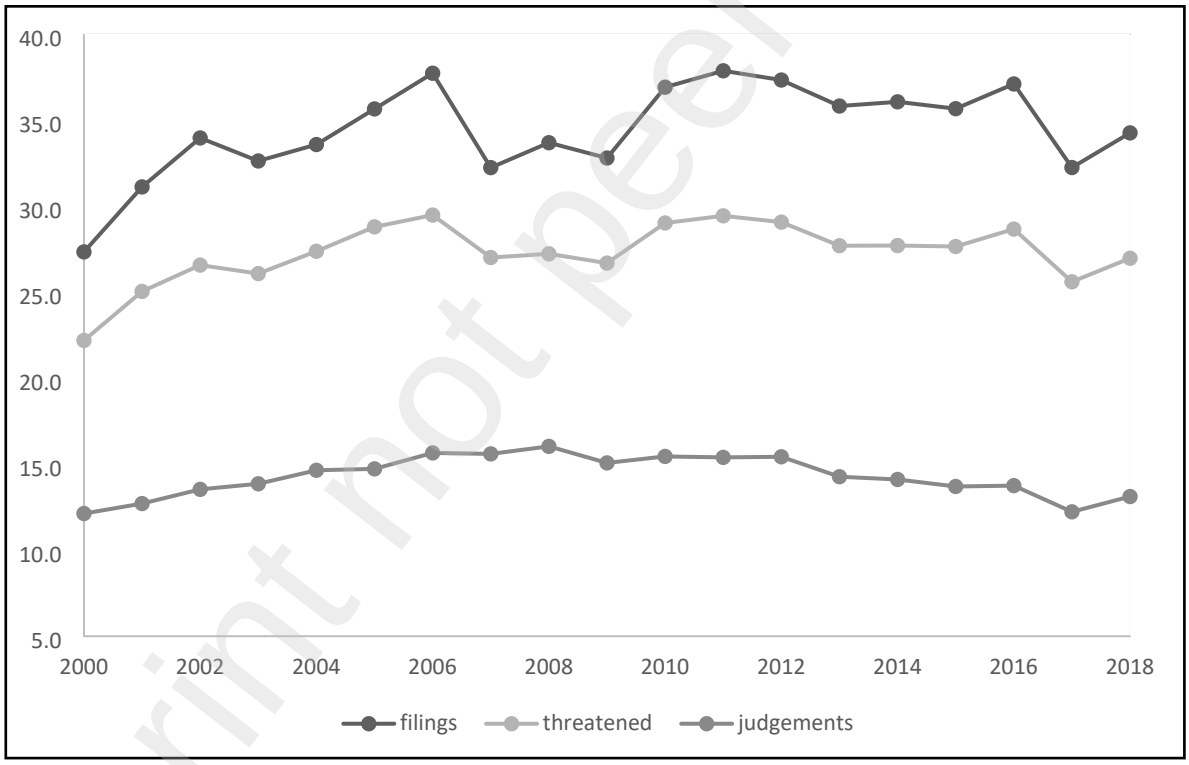
alleged violation of the lease terms, usually for non-payments.[9] Before the eviction, tenants receive a notice explaining the reason for terminating the lease and the time until the lease ends. Tenants must respond by answering the court summons or appearing in court. If the tenant fails to respond, a default judgment may be issued in favor of the landlord. If the tenant responds, there is the possibility that the landlord and the tenant reach an agreement that stop the eviction. To avoid eviction, tenants must pay late fees, court costs, and past-due rent. Leung et al. (2020) estimate that each eviction filing costs tenants on average around $180 in fines and fees, raising monthly housing costs by 20%. They also show that in 2014 nearly one-third of households facing eviction had multiple filings from the same address.

Figure 1: Number of filings, threatened and judgement cases per tract, 2000-2018



Note: According to Eviction Lab, eviction "filings" represent all cases filed in an area, which can result in multiple cases filed for the same address in a year. The eviction "threatened" status corresponds to households that received at least one filling, and "judgements" correspond to those filings that result in eviction judgments. The figure contains the mean year average per tract for all US census tracts.

[9] Collinson et al. (2022) explains that according to "*the 2013 American Housing Survey, 75 percent of households who reported being threatened with an eviction reported that the reason for the threat was failure to pay rent*."

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Court judgment in eviction cases can resolve in favor of either the landlord or the tenant. Hearings are usually brief (Doran et al., 2003; Desmond, 2016), with tenants often unrepresented while landlords are usually represented by an attorney (Collinson et al., 2022; Cassidy and Curry, 2023). If the tenant fails to address the issue that resulted in the lease violation and the landlord is successful at the hearing, the judge issues a "writ of restitution", authorizing the local law enforcement agency to evict the tenant (Desmond, 2016; Falcone, 2022). After an eviction judgment, families can be forced to vacate the property in a short period of time. In some occasions, sheriff deputies remove families and their possessions are piled on the curb or confiscated by movers (Desmond and Kimbro, 2015). To find a new place to stay can be difficult, since future landlords can see eviction cases on tenant screening reports (Gromis et al., 2022).

Figure 1 shows the evolution of evictions over the period 2000-2018 for all US census tracts. The variable *threatened* is the number of Households Threatened with Eviction, and is the main dependent variable in our analysis. The variable *filings* is the sum of all evictions filings, while the variable *judgements* measures the number of eviction filings that resulted in eviction judgments. According to Gromis et al. (2022), between 2000 and 2018, landlords in the U.S. filed over 3.6 million eviction cases annually, affecting nearly 7% of renting households. Informal evictions, which can be two to three times more frequent than formal ones, further increase this figure (Desmond, 2016; Falcone, 2022). Although the filing rate has decreased in the last years, this is in part due to the expansion of the renter population and to changes in renters' socio-economic profiles.

Eviction filings are widespread throughout the US, with the highest rates in the in the Southeastern states (see Figure 2 for an illustration). According to Gromis et al. (2022), "*population, median rent, and rent burden showed only modest, positive bivariate associations with county-level filing rates in our data, while poverty rates showed no bivariate association*". They also show that states that do not require eviction notices before filing for nonpayment have significantly higher eviction rates.

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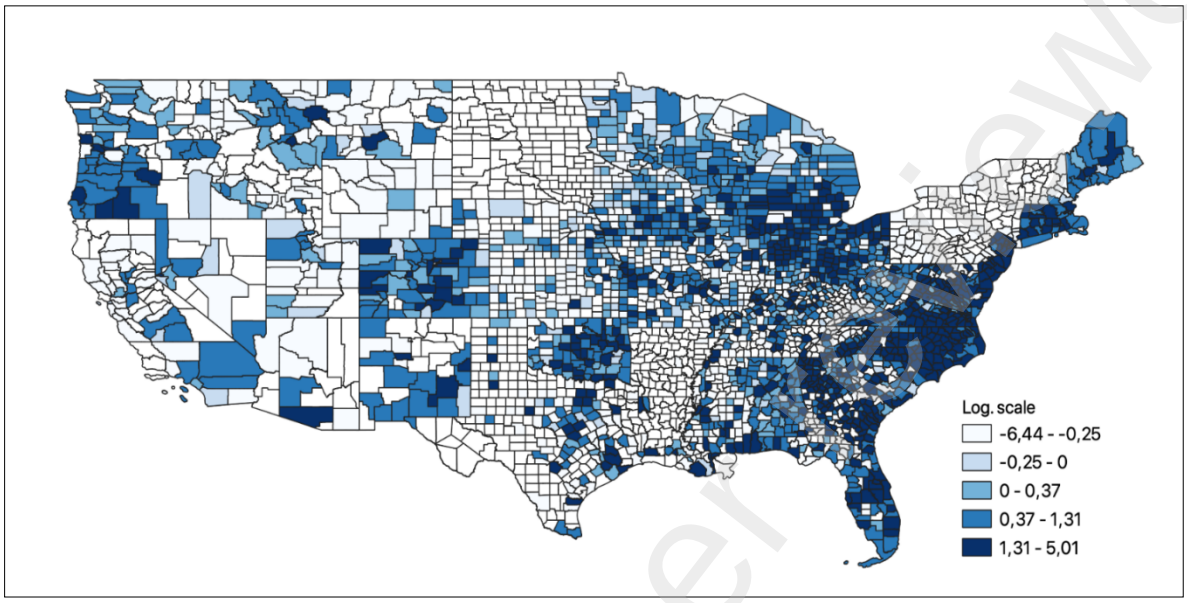
Figure 2: Households threatened with eviction, at the county level: 2018



Note: The figure shows the intensity of the evictions at the county level, using the original dataset of Evictions Lab (EL). The analysis shows the log of the variable *Households threatened with eviction,*

4. Branch consolidation

In 1994, the Riegle-Neal Act initiated a significant shift in the US banking regulation by removing strict barriers to interstate banking and branching. Prior to the Act, most states restricted banks from expanding their operations across state lines. The Act aimed to increase efficiency and allowed bank holding companies to acquire banks nationwide and diversify their assets. This led to a wave of mergers and acquisitions, with many small and mid-sized banks being absorbed by larger, geographically diverse banks. Initially, some states resisted the new regulation, adopting measures to protect local banks, but many of these restrictions soon disappeared, and the banking market continued its consolidation.

In 1999, the Gramm-Leach-Bliley Act promoted the integration of the commercial and investment banking, leading to greater competition and more sector consolidation through mergers and acquisitions (Chaddock, 1994; Vives, 2016). As a result, the number of banks significantly declined, with smaller entities being acquired by larger ones seeking economies of scale. Between 1993 and 2000, the number of banks decreased by around 25%, falling to

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approximately 10,000. Despite this consolidation, the number of bank branches expanded, peaking at nearly 100,000 offices in 2009.

The reduction in the number of banks intensified after the 2007-08 economic crisis. This was not due to an increase in bank failures, but rather a result of the decline of bank entries and the continued pace of mergers and acquisitions. From 2009 to 2013, new bank entries almost halted, and branch expansion into new geographic markets dropped dramatically. This can be attributed to low bank profitability and the implementation of new banking regulations.

Following the financial crisis of 2007, a range of measures were implemented to safeguard the banking sector from future crisis. The 2010 Dodd-Frank Wall Street Reform and Consumer Protection Act brought significant regulatory changes. Section 613 of the Act partially dismantled the restrictions on de novo branching by out-of-state banks, allowing them to open new branches more easily. The Act also promoted transparency and accountability in banking, with the goal of preventing future financial crises. It established measures to avoid bank bailouts, monitor risks in the financial system, and reform the securities market, regulation of derivatives, and rating agencies. Additionally, the Act tried to reinforce the stability of the banking system by introducing the Volcker Rule, which prohibited banks from engaging in proprietary trading and restricted their investments in hedge funds and private equity funds. The Act also mandated stress tests and imposed higher capital and liquidity requirements to enhance banks' resilience to future economic shocks (Barth et al., 2013).

As a result of this consolidation process, the number of bank branches declined to approximately 85,000 locations by 2020. Calzada et al. (2023) show that mergers involving large banks played a significant role in the adjustment of branch density. Interstate mergers of large banks led to a 5% decrease in branch density in tracts exposed to these mergers, while the impact of intrastate mergers was even more pronounced. Large mergers reduced branch density by 14% in rural tracts and by 12% in tracts with overlapping branch networks. This debranching trend is also associated to the widespread adoption of online banking, which led to important cost savings compared to brick-and-mortar branches.

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5. Data

Our paper combines a dataset on evictions obtained from the Evictions Lab (EL) at Princeton University and information on bank branches obtained from the Federal Deposit Insurance Corporation (FDIC). We complete our dataset with information on socio-demographic from the US Census Bureau. As a result, we obtain a dataset for 73,057 census tracts located in 3,145 counties in the 50 states of the United States and the District of Columbia for the period 2000–2018.[10]

Data on evictions. Our dataset on evictions has been obtained from the Evictions Lab (EL) at Princeton University, which is the most complete data set on evictions in the United States (Desmond et al. 2018; Gromis et al. 2022a and 2022b).[11] EL has collected court records from several states. However, due to the lack of centralized eviction data in many states and some states' unwillingness to release this information, EL also purchased more comprehensive datasets of public eviction records from two private companies, LexisNexis Risk Solutions and American Information Research Services Inc. The dataset used in this study includes residential eviction cases filed between 2000 and 2018 across 50 states and the District of Columbia (DC). It accounts for eviction cases involving single individuals, families, or multiple families within a single residential unit, and aggregates this data at the census tract and calendar year level. Specifically, EL geocodes tenant addresses from court cases and aggregates the data at the census-tract level based on 2010 Census boundaries. Foreclosures, evictions of commercial tenants, and forced moves from public structures are excluded from the dataset. A limitation of the dataset is that is does not reflect informal evictions, which can occur two or three times more frequently than formal evictions (Desmond, 2016). Moreover, court records do not show whether tenants ultimately left the property at the end of the case. Indeed, the tenant and the landlord can reach an agreement to resolve the issue.

The main variable of interest of our analysis is *Households Threatened with Eviction*, which measures the number of households in the census tract-year that at least received one eviction filling. It only considers the first (or only) eviction filling of each household in the

[10] The number of tracts defined by the US Census increased in the period examined; according to our dataset, there were 65,443 tracts in the 2000-2009 decade and 73,057 in the 2010-2019 decade.
[11] All data related to evictions has been collected from the Eviction Lab (EL). https://evictionlab.org/ . For a detailed description see: ttps://evictionlab.org/docs/Eviction%20Lab%20Methodology%20Report.pdf

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tract-year.[12] In section 8.4, we perform a robustness check that consider the variable *Eviction Filings*, which represents the total number of eviction cases at the census tract-year level. This includes instances of multiple fillings ("serial fillings"), where a landlord issues a series of filings against the same household at the same address. We also use the variable *Judgement*, which considers the number of eviction filings resulting in eviction judgments.

Data on bank branches. We have obtained the data on bank branches from the Summary of Deposit (SOD) and the Reports of Structure Changes (ROC) provided by the Federal Deposit Insurance Corporation (FDIC). The FDIC compiles SOD data for all insured financial institutions, including commercial banks and savings and loans associations. This dataset offers a comprehensive record of each institutions' financial status, including assets, deposits, and other relevant information, such as the type of financial services offered and the bank's location. The data is collected at the branch level, providing a detailed view of the characteristics of the branches. A useful feature of the SOD data is the inclusion of latitude and longitude coordinates for each branch. This has allowed us to use the Geographic Information System (GIS) software to map the branches onto census *tracts* and calculate the distances from the centroids of each tract's census *blocs* to the branches.

The ROC data provides valuable insights into the non-financial activities of financial entities. This includes information on institutional and structural changes such as mergers, failures, and the establishment of new offices. To conduct our analysis at the census tract level, we aggregate the local branch information. This allows us to calculate various metrics, such as branch density, inter- and intra-state mergers, and bank entries and exits for each census tract.

Our empirical analysis uses two variables to measure "financial exclusion". First, the variable *Distance* measures the distance of the population to the nearest bank branch at the tract-year level. Specifically, this variable computes the average distance from the centroids of the census blocks of each tract to the nearest bank branch, weighting for the relative population of the census blocks. We have processed this information with GIS software to obtain the

[12] As explained by Gromis et al. (2022b), the number of fillings only offers a partial view of the eviction prevalence and its consequences. "*First, court records do not provide information about whether tenants vacated the rental property before, during, or after an eviction filing. When multiple cases were filed against the same household, we knew that tenants remained at the property after the previous filing but did not know whether the case series ultimately resulted in displacement. For households who only appeared once in the data, we had no means of assessing whether the single filing resulted in displacement*."

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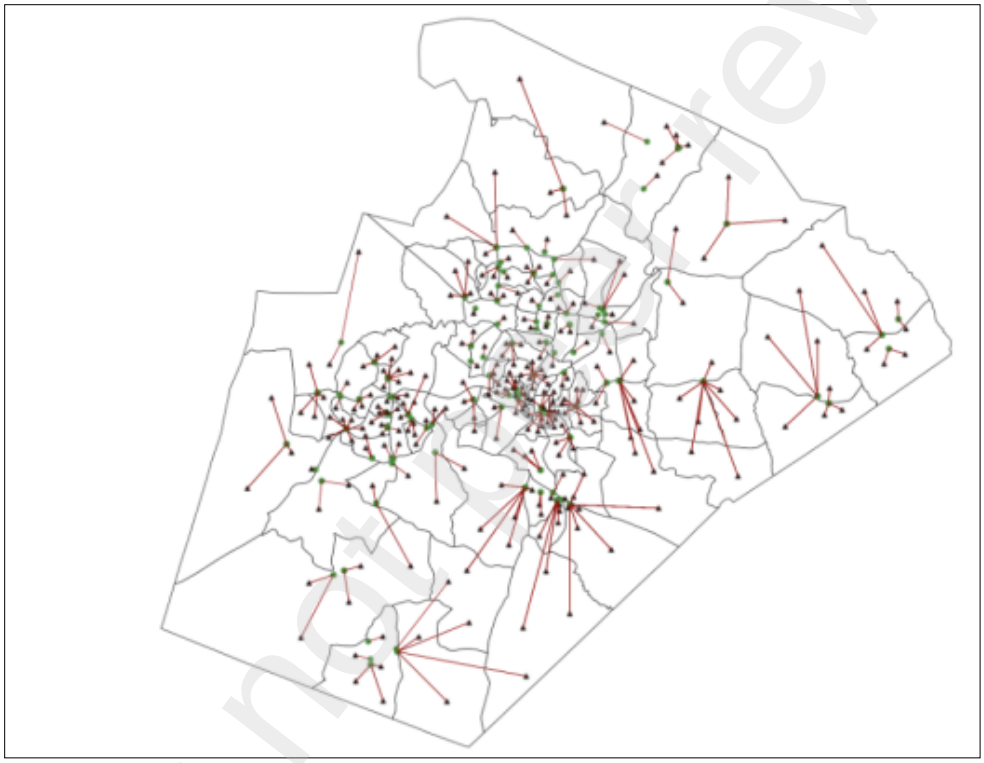
Euclidean distance from the centroid of the census blocks to the branches (hence, we assume that all the population of the blocks is located at their centroid). We exclude from our sample those tracts that do not have a bank branch. Figure 3 illustrates how we have calculated the variable *Distance*, using as an example the tracts in the Wake Cunty (NC), in 2003. The red lines show the distance from the centroids of the blocs (black triangles) to the nearest bank branch (green dots).

Figure 3. Distance between bank branches and census blocks, Wake County (NC), 2003.



Note: The green dots show the branch location and the black triangles the census blocks' centroids. The red lines represent the distance from the blocks' centroids to the closest bank branch (Euclidean distance, measured in meters). For illustrative purposes, we use the same county than Nguyen (2019),

Second, the variable *Branches per Inhabitant* reflects the availability of bank services in the neighborhood. This variable is calculated at the tract level and accounts for differences in the population size of the tracts.

The variables *Distance* and *Branches per Inhabitant* are useful to identify changes in "financial exclusion". A greater average distance to bank branches implies reduced access to banking services. In this sense, branch closures within a census tract might entail an increase

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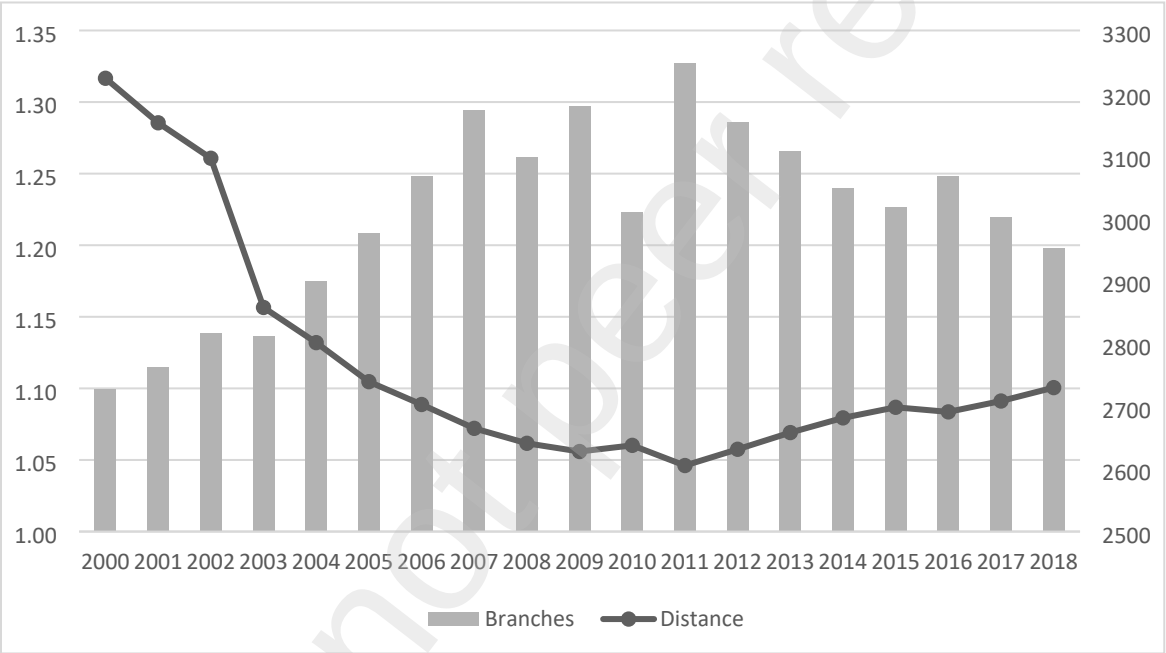
in the distance that an average household must travel to access its local branch and obtain financial assistance. It is important to note that the number of branches in each census tract is influenced by local market competition, the banks' commercial strategies, and the characteristics of the population. However, the number of branches can also be exogenously affected by mergers involving large banks. Figure 4 shows the evolution of the variables *Distance* and *Branches per Inhabitant* in the period examined.

Figure 4: Distance to the closest branch and branches per inhabitant, 2000-2018



Note: This figure plots the evolution in the average number of branches per tract (left axis) and the evolution in the average distance to the closest bank branch (right axis).

Data on socio-economics and demographics. Our analysis incorporates socio-economic data and median family income at the census tract level, obtained from the US Census Bureau. For the period 2000-2008, we rely on information from the 2000 Decennial Census, and for the period 2009-2018, we use data from the American Community Survey (ACS).[13] The Census also provides information on the land area of each tract, which we use to calculate population densities. The variable *Density Population* is calculated by dividing the

[13] https://www.census.gov/programs-surveys/acs/data.html

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population of a tract by its area. Other variables considered in the analysis include median family income, which provides insights into the economic well-being of the community, the percentage of the population over 60 years old (% older), the percentage of the population with a college degree (% higher education), the percentage of the population belonging to minority ethnic groups (% minority), and the percentage of rental households in the tract (% rental households).

An essential element in our analysis is to identify the effects of financial exclusion on census tracts with heterogeneous geographical characteristics. In particular, we examine the existence of different effects across urban and rural populations. The American Community Survey (ACS) data from 2018 shows that close to 19% of the US population lived in rural regions, which constituted around 97% of the country's entire land area. The distinction between urban and rural populations predominantly relies on population density. In our paper, we designate tracts as urban if more than 50% of the residents live in urban areas. Our results remain very similar when we use a striker threshold, defining urban tracts as those were over 75% of the population resides in urban settings, although this significantly reduces the number of tracts classified as rural. Based on these criteria, our sample includes 15,343 urban tracts and 1,568 rural tracts that have been impacted by mergers at some point. It is important to note that in the recent years, large banks have increasingly focused their commercial activities in urban centers, leaving small and community banks to serve the slower-growing and less lucrative rural markets. This situation has enabled smaller banks to maintain their presence in rural communities.

Table 1 presents the descriptive statistics of the variables used in our empirical analysis. All variables exhibit a high degree of variability across tracts, given that the standard deviation is high in relation to the mean. There are remarkable differences between urban and rural tracts. The average number of evictions is much higher in urban than in rural tracts, suggesting that evictions is primarily an urban phenomenon, despite the higher density of bank branches in these areas. At the same time, the change in branch density is higher for urban tracts, as they have been more affected by the consolidation process in the banking market. As expected, urban tracts tend to be denser, wealthier, and have a higher percentage of college graduates. Moreover, they have a higher percentage of minorities and renters.

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Table 1. Descriptive statistics of the variables used in the empirical analysis

	All sample		Urban tracts		Rural tracts	
Variable	**Mean**	**Std. dev.**	**Mean**	**Std. dev.**	**Mean**	**Std. dev.**
Evictions	25.09	45.22	29.81	48.43	6.12	7.87
Distance	2607.75	6639.88	901.97	761.69	5471.43	5031.17
Branches per inhabitant	7.36	82.18	8.08	95.83	5.65	17.65
Median Income	51909.29	25639.83	54028.84	27651.78	43510.71	14741.53
Population Density	4018.58	10042.58	5474.44	11428.02	61.99	112.71
% minorities	0.20	0.21	0.24	0.22	0.09	0.14
% college graduates	0.27	0.17	0.30	0.18	0.16	0.08
% households with individuals above 60 yo	0.33	0.11	0.32	0.12	0.38	0.08
% houses occupied by renters	0.34	0.21	0.39	0.22	0.20	0.07
Intercounty mergers	0.06	0.28	0.07	0.31	0.01	0.15
Interstate mergers	0.34	0.812	0.42	0.90	0.07	0.32

6. Empirical Strategy

The objective of our empirical analysis is to estimate the causal relationship between financial exclusion and evictions in the US from 2000 to 2018. Our analysis in Section 4 has shown that the number of bank branches increased until the 2008 financial crisis, but that it faced a significant decrease thereafter. We hypothesize that these changes in brick-and-mortar branches at the census tract level could had affected the financial support available to tenants, influencing their capacity to pay their rents. To analyze the relationship between the financial exclusion and evictions we estimate the following baseline model:

$$Households\ Evictions_{cy} = \beta\ FE_{cy} + \gamma X_{cy} + Tract_c + Year_y + u_{cy}, \quad (1)$$

where $Households\ Evictions_{cy}$ is the number of households threated with an eviction in the census tract c in year y, and FE_{cy} is a measure of *Financial Exclusion*. In our baseline model, we measure financial exclusion with the variable *Distance*, which reflects the average distance of the population to the bank branches in tract c in year y. In the robustness analysis of Section 8.3 we use the variable *Branches per Inhabitant* as an alternative measure for financial exclusion. This variable reflects the branch density in the census tract c in year y.

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Moreover, X_{cy} is a set of control variables varying across tracts and years, and $Tract_c$ and $Year_y$ are census tract and year fixed effects that control for idiosyncratic shocks at the tract and year level. The usual iid assumption applies to the error term u_{cy}.

All variables without bounded values are transformed in logarithms. This change mitigates the severe skewness that affects some variables, particularly the dependent variable. It also reduces the influence of outliers and helps to interpret the coefficients in terms of elasticities.

The main challenge facing the analysis of the causal impact of financial exclusion on evictions is the potential presence of unobservable factors at the census tract level that can be correlated with both branch density and the socio-economic conditions of the population. To address this potential endogeneity problem, we pursue an instrumental variable (IV) identification strategy. For this, we need some variable (instrument) that is correlated with the population's distance to bank branches but has no direct effect on the evictions, except through its influence on branch proximity. Following Nguyen (2019) and Calzada et al. (2022), we use large bank mergers during this period as our instrument. These mergers could directly modify the number of branches available in a tract but represent exogenous shocks to households potentially affected by evictions. We focus on mergers involving banks with at least $1 billion in pre-merger assets and where the acquired bank operated more than 80 branches. This limitation results in 106 mergers in which participated 153 large banks in the period 2000–2018, affecting 16,687 tracts belonging to 2,127 counties. Large bank mergers often pursue multiple objectives, such as market expansion, cost reduction, and increasing market influence. However, it is unlikely that they are driven by the specific socio-economic circumstances that affect every local market. To further ensure that this is indeed the case, we exclude from our analysis "intra-county mergers" where both the acquiring and the acquired banks share the same county headquarters. As a result, our final sample includes 86 "interstate mergers" between banks from different states and 20 intrastate/inter-county mergers" between banks from the same state but headquarters in different counties.

Following mergers, it is common for the newly formed banks to undergo a period of consolidation, which often involves closing branches in areas where the previously separate networks had a presence. As a result, tracts with branches of the banks participating in mergers are more likely to experience post-merger closures.

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The instrumental variables (IV) version of equation (1) replaces *FE* by *Financial Exclusion*, where *Financial Exclusion* is the predicted value of *FE* from the following first-stage equation,

$$Financial\ Exclusion_{cy} = \theta_0 + \theta_1 Mergers_{cy} + \theta_2 X_{cy} \ + \ \omega_{cy}. \qquad (2)$$

The instruments of the variable *Financial Exclusion* are the variables *Intercounty Mergers* and *Interstate Mergers* which reflect the number of times a tract has been exposed to intrastate/intercounty and interstate mergers, respectively. Notice tracts could be exposed to mergers several times during the period examined. We consider that a tract was exposed to a merger when the acquiring or the acquired bank was operating in the tract before the merger.

The merger variables are instruments for the independent variable $Distance_{cy}$ under the assumption that cov(Mergers, u_{cy})=0. This means that mergers are orthogonal to changes in the number of evictions in a census tract. That is, we assume that large banks do not choose their merger strategy considering the evolution in the evictions.

7. Results

7.1 Main Results

This section analyses the effects of financial exclusion on evictions. Table 2 presents the results of the regressions that analyze the effect of the variable *Distance* on the number of *Households Evictions* at the census tract level, using the ordinary least squares method (OLS). Columns (1) and (2) use the entire sample, columns (3) and (4), the sample of urban tracts, and columns (5) and (6) the sample of rural tracts. All regressions include the same set of control variables*: median income, population density, percentage of minorities, percentage of college graduates, percentage of households with individuals above 60 years old, and percentage of houses occupied by renters*. They also include tract and year fixed effects. Standard errors are clustered at the tract.

The OLS analysis reveals a positive and significant effect of the variable *Distance* on evictions, except in rural tracts. Specifically, column (1) shows that for the full sample a 1%

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increase in the distance leads to a 0.27% increase in the number of threatened households. Column (3) shows that this effect rises to 0.33% in urban tracts, and column (5) shows that distance does not have a significant effect in rural tracts. In addition, columns (2) and (4) show that *Interstate Mergers* have a positive and significant impact on evictions when considering both the full sample and the urban subsample. This is consistent with the descriptive statistics in Table 1, which show that bank mergers mainly affect urban tracts. We also find that *Intercounty Mergers* have a negative effect on rural tracts, although this result is weakly significant.

A potential limitation of this analysis is that banks do not randomly decide which branches to close. For example, they can close branches in those neighborhoods with lower profitability, due to changes in the economic activity or in the socio-economic composition of the population. In order to deal with this endogeneity problem, we next pursue a two-stage least squares instrumental variable estimation (2SLS-IV). We use the intercounty and interstate mergers of larger banks as instruments for the variable *Distance* (see a similar approach in Nguyen, 2019 and Calzada et al. 2023).

Table 2. Effect of distance on evictions (OLS regressions)

	Evictions					
	(1)	(2)	(3)	(4)	(5)	(6)
Intercounty Mergers		0.0164		0.0258		-0.0801*
		(0.0162)		(0.0183)		(0.0481)
Interstate Mergers		0.0588***		0.0672***		-0.000722
		(0.00599)		(0.00634)		(0.0242)
Distance	0.0274***		0.0334***		-0.00804	
	(0.00713)		(0.00845)		(0.0131)	
Sample	All	All	Urban	Urban	Rural	Rural
Period	2000-2018	2000-2018	2000-2018	2000-2018	2000-2018	2000-2018
R-squared	0.046	0.046	0.048	0.049	0.043	0.043
Number of tracts	58,072	58,072	45,542	45,542	9,074	9,074
Observations	751,599	751,599	613,516	613,516	90,661	90,661

Notes: Robust standard errors, in parenthesis, are clustered at municipality level *** p<0.01, ** p<0.05, * p<0.1. In columns (1), (3) and (5) the main explanatory variable is *Distance*, which is constructed as the weighted (by population) average of the distance (by meters) within the tract´s blocks to the closest banking office. In columns (2), (4) and (6) the main explanatory variables are Intercounty Mergers and Interstate Mergers. All regressions include year and tract fixed effects.

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Panel A of Table 3 presents the results of the first stage of the IV estimation and panel B the results of the second stage. Columns 1 uses the full sample, while columns 2 and 3 focus on urban and rural tracts, respectively. The results in panel A shows that the two types of mergers (intercounty and interstate) have the expected positive and significant effect on the distance between the population and the bank branches. Furthermore, the F-test of significance of the instruments in the first stage is higher than the threshold of 10 for the full sample and the subsample of urban tracts, which is the commonly accepted rule of thumb for identifying strong instruments. In addition, the Hansen test confirms that the instruments are exogenous in all cases. These results imply that in the period studied mergers had a negative effect on financial inclusion, as they increased the distance consumers had to travel to access banking services.

Column 1 in panel B shows that, when considering the full sample, the distance to bank branches has a positive and significant impact on the number of households threated with eviction. Specifically, we find that a 1% increase in distance led to a 2.39% increase in the number of households filled with an eviction. Column 2 focuses on urban tracts and indicates that a 1% increase in distance led to a 3% increase in evictions. In contrast, we do not find a statistically significant relationship between distance and eviction when examining rural tracts.

Our results confirm that the effect of distance on evictions is mainly concentrated in urban tracts, which have a higher branch density and are more affected by mergers. To simplify the exposition of our results, the rest of our analysis will focus on urban tracts.

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Table 3. Effect of distance on evictions (IV regressions)

		Distance	
Panel A: First stage	(1)	(2)	(3)
Intercounty Mergers	0.025***	0.019***	0.106***
	(0.005)	(0.005)	(0.028)
Interstate Mergers	0.022***	0.021***	0.043***
	(0.001)	(0.001)	(0.011)
		Evictions	
Panel B: Second stage	(1)	(2)	(3)
Distance	2.391***	3.005***	-0.422
	(0.291)	(0.362)	(0.352)
Sample	All	Urban	Rural
Period	2000-2018	2000-2018	2000-2018
F-test joint sign. (2nd stage)	370.77***	300.72***	66.55***
F-test instruments (1st stage)	94.74***	73.96***	16.38***
Hansen test (1st stage)	1.784	1.697	0.927
Number of tracts	57,053	44,984	8,669
Observations	750,580	612,958	90,256

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$. The results in Panel A correspond to the first stage of the IV regressions, where the instruments for the variable *Distance* are *Intercounty* and *Interstate Mergers*. Results in Panel B correspond to the second stage of the IV regressions. All regressions include tract and year fixed effects.

7.2. The impact of the financial crisis

Our description of the evolution of the US banking system in Section 4 has identified two clear trends. First, between 2000 and 2008 the number of bank entities decreased at an average annual rate of 2.2%. Despite this consolidation, the size of the bank branch networks continued to expand, peaking at 99,550 branches in 2009. This expansion was partially driven by the Riegle-Neal Act, which allowed banks to enter new markets, reduce their costs, and diversify assets. Many banks implemented this strategy through mergers and acquisitions (McLaughlin, 1995; Cohen and Mazzeo, 2010; Aubuchon and Wheelock, 2010; Aguirregabiria et al. 2016; Gropp et al., 2019; Calzada et al. 2022). Our hypothesis is that this expansion in the number of bank branches may had reduced financial exclusion and, consequently the number of evictions. Indeed, those tracts that benefited with an increase in bank branches likely had an improvement in the quality of their financial services.

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Second, after 2008, the number of banks decreased at an average annual rate of 4,2%, and the number of branches shrank by 1.3% per year, falling to 85,050 branches by 2020. Interestingly enough, during the 2008-2020 period, the US population grew by 8.5%, resulting in a 26.5% increase in the average number of people served per branch—from 3,000 people per branch in 2008 to 3,800 in 2020. Following the 2007–2008 financial crisis, the Federal Reserve's policy of maintaining near-zero federal funds rates pushed lending rates down, reducing the bank's net interest margin. This, coupled with the fall in demand (e.g. loans and deposit-taking services) and the implementation of new banking regulations (e.g. Dodd-Frank Act and FDIC regulations), reduced bank profitability and led to mergers aimed at reducing the operating costs and improving efficiency (DeYoung et al. 2009). As a result, the number of bank branches decreased significantly.

We hypothesize that census tracts experiencing larger reductions in the number of branches were more likely to see an increase in households threatened with eviction. We also expect that this effect was more important in urban tracts, as they had a higher density of branches. In addition, it could be considered that urban populations have better access to broadband Internet than those living in rural areas and, therefore, are more likely to bank on-line.

Table 4 presents the same analysis as in Table 3 but divides the sample into three periods: 2000-2007 and 2008-2013 and 2014-2018. The first interval corresponds to a period of expansion of bank branches, while the other two reflect a period of market consolidation. Our results show a positive and significant effect of distance to branch services on evictions during the 2000-2007 and 2008-2013 periods. In particular, a 1% increase in distance to bank branches resulted in a 5% and 1.6% increase, respectively, in the number of households filled with an eviction. These results suggest that during the period of branch expansion, an increase in the availability of bank branches may have helped tenants to better manage their finances, reducing the likelihood of evictions. Likewise, after the financial crisis branch closing could exacerbate tenants' financial difficulties and make them more exposed to evictions.

In contrast, we do not find a significant relationship between distance and evictions for the 2014-18 period. To this point, note that the IV analysis for this period is not valid, as the instruments are neither strong (the F-test in the first stage is less than 10) nor exogenous (the Hansen test rejects that the instruments are exogenous). This could be explained by the

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relatively small number of interstate and intrastate mergers in this period, which make these variables ineffective as instruments. This is a caveat that must be considered in the interpretation of the results.

Table 4. Effect of distance on evictions (IV regressions). Subsamples by time period

		Distance	
Panel A: First stage	(1)	(2)	(3)
Intercounty Mergers	0.0189***	0.026***	0.016**
	(0.004)	(0.010)	0.007
Interstate Mergers	0.015***	0.022***	0.002
	(0.002)	(0.002)	(0.005)
		Evictions	
Panel B: Second stage	(1)	(2)	(3)
Distance	5.058***	1.693***	-2.491
	(0.813)	(0.489)	(2.213)
Sample	Urban	Urban	Urban
Period	2000-2007	2008-2013	2014-2018
F-test joint sign. (2nd stage)	377.19***	66.78***	218.71***
F-test instruments (1st stage)	36.14***	33.34***	2.42*
Hansen test (1st stage)	0.175	0.0002	12.171***
Number of tracts	31,553	40,116	40,758
Observations	217,180	204,421	187,588

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$. The results in the top-part of the table correspond to the first stage of IV regressions where the instruments for distance are the Intercounty Mergers and Interstate Mergers. Results in the bottom-part of the table correspond to the second stage of IV regressions.

7.3. Heterogeneous effects of the financial exclusion

We next investigate the heterogeneity of the effects of financial exclusion by considering different classifications of the census tracts in our sample. Specifically, we consider the socio-economic and demographic characteristics of the tracts, such as median family income, population density, percentage of the population over 60 years old, percentage with a college degree, percentage belonging to minority ethnic groups, and percentage of rental households. Tracts are classified based on whether they are above or below the mean of the value of each of these variables (results are similar when using the median). Table 5 repeats the IV estimations

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of Table 3 for each of the subsamples. For simplicity, we do not report the results of the first-stage regressions, but the results of the Hansen test show that the instruments are exogenous, except in tracts with high income, low population density and low percentage of renters. The results from the second-stage regressions confirm the significant effect of financial exclusion on the number of evictions. Interestingly enough, we find a larger impact in tracts with a higher percentage of minority ethnic groups and a higher percentage of renters.

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Table 5. Effect of distance on evictions (IV regressions). Subsamples by tracts characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Distance	2.990***	2.000***	3.066***	1.038***	3.210***	0.616*	3.176***	1.819***	1.674***	2.580***	3.248***	1.681***
	(0.465)	(0.414)	(0.408)	(0.367)	(0.454)	(0.331)	(0.483)	(0.317)	(0.358)	(0.460)	(0.500)	(0.304)
Sample	High income	Low income	High pop density	Low pop density	High % minorities	Low % minorities	High % college	Low % college	High % above 60	Low % above 60	High % renters	Low % renters
F-test joint sign. (2nd stage)	175.75***	220.33***	235.44***	225.61***	310.24***	159.82***	219.01***	73.53***	177.31***	72.16***	215.70***	185.82***
Number of id	38,005	32,090	32,336	26,638	34,776	31,403	35,617	31,134	41,220	35,648	34,188	30,726
Observations	372,633	374,282	443,923	306,012	412,770	334,614	377,378	369,678	368,772	376,376	419,970	327,625

Notes: In columns (1), (3) and (5) the main explanatory variable is Distance that is constructed as the weighted (by population) average of the distance (by meters) within the tract´s blocks to the closest banking office. In columns (2), (4) and (6) the main explanatory variables are Intercounty Mergers and Interstate Mergers. All regressions include tract and year fixed effects.

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8. Robustness checks

This section presents several robustness checks to validate our results. First, we estimate the model in equation (2) considering a matching procedure to account for possible observable differences between tracts exposed and unexposed to mergers that could affect the results. Second, we re-estimate the model focusing on mergers where the acquirer and the acquired banks had overlapping branches within the same tract. Third, we use as an alternative measure of financial exclusion the number of *branches per inhabitant* at the census tract level. Finally, we consider two alternative variables to measure the exposition to evictions, the *total number of evictions* at the census tract and the *number of eviction filings resulting in eviction judgments*.

8.1 Matching estimation

Our previous analysis has analyzed the effect of distance on evictions using as exogenous source of variation the exposure to mergers of large banks. Even though the Hansen test suggests that the mergers are valid exogenous instruments, in this section, we apply a matching procedure to control for potential pre-existing observable differences between tracts exposed and unexposed to mergers that might influence our results. Matching techniques help eliminate selection biases by pairing similar tracts affected and unaffected by mergers.

We apply a propensity score matching procedure to build a subsample of unexposed tracts with similar income to those of the tracts exposed to the mergers. Specifically, we impose that tracts in both groups are comparable in terms of their median income across all years of the analysis. This allow us to control for potential socio-economic distortions caused by the financial crisis.

Table 6 shows the same regression than in column 2 in Table 3 but using the matched sample. The results confirm a positive and significant impact of distance on evictions, with a slightly higher coefficient compared to the unmatched sample. When we consider the entire period and urban tracts, we find that a 1% increase in distance leads to a 3.48% increase in the number of households evicted. Therefore, our baseline result in Table 3 - that increased distance to bank branches leads to more households threatened by eviction – does not seem to be affected by income differences between tracts affected and not affected by mergers.

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Table 6. Effect of distance on evictions: matched sample by income

Panel A: First stage	Distance
Intercounty Mergers	0.015***
	(0.005)
Interstate Mergers	0.016***
	(0.001)
Panel B: Second stage	Evictions
Distance	3.487***
	(0.514)
Sample	Urban & matched
Period	2000-2018
F-test joint sign. (2nd stage)	120.58***
F-test instruments (1st stage)	44.96***
Hansen test (1st stage)	1.616
Number of tracts	35,194
Observations	251,664

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** p < 0.01, ** p < 0.05, *p < 0.1. Standardized % bias across income. Unmatched (-5.3), Matched (1.9). All regressions include tract and year fixed effects.

8.2 Effects of mergers in tracts with overlapping branches

This robustness analysis repeats the estimation of column (2) in Table 3, but focusing on mergers where prior to the transaction the acquirer and the acquired banks had overlapping retail branch networks within the same tract. We expect that in tracts with overlapping bank branches, the merged entity will be more likely to close branches. The results in Panel A of Table 7 confirm a positive and significant impact of intercounty mergers on distance, with a higher coefficient than those in Table 3. However, we do not find a significant effect of interstate mergers on the distance to the branches. Panel B shows that the average distance to bank branches in the tract has a positive and significant effect on number of households threatened with eviction. Specifically, a 1% increase in distance led to a 2.79% increase in the number of households filled with an eviction.

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Table 7: Effect of distance on evictions: overlapping branches

Panel A: First stage	Distance
Intercounty Mergers (overlap)	0.024***
	(0.009)
Interstate Mergers (overlap)	0.0171
	(0.015)
Panel B: Second stage	Evictions
Distance	2.790**
	(1.403)
Sample	Urban
Period	2000-2018
F-test joint sign. (2nd stage)	313.90***
F-test instruments (1st stage)	15.94***
Hansen test (1st stage)	0.381
Number of tracts	44,984
Observations	612,958

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$. The results of the table correspond to the first stage of IV regressions where the instruments for distance are the Intercounty Mergers and Interstate Mergers with overlap. The regression includes tract and year fixed effects.

8.3 Alternative measure of branch closures

This section uses the number of *Branches per Inhabitant* at the census tract level as a measure of financial exclusion. This variable shows the availability of financial entities in the tenants' neighborhood, complementing our previous analysis. Panel A in Table 8 presents the results of the first stage of the IV estimation. We find that the variables *Intecounty* and *InterState Mergers* have a negative and statistically significant effect on the number of branches per inhabitant. Indeed, mergers spur bank closures, with the consequent reduction in branch density. The value of the F-test for the first stage shows that the instruments for the IV estimation are strong, and the Hansen test shows that the instruments are exogenous.

Results of the second stage of the IV estimation have a similar interpretation than those in the previous analysis. A 1% increase in the number of branches per inhabitant lead to a 0.67% reduction in the number of households threatened with eviction. This confirm that the availability of bank branches in the tenants' neighborhood plays a significant role in reducing evictions.

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Table 8. Effect of branch density on evictions (IV regressions)

Panel A: First stage	Branch density
Intercounty Mergers	-0.073***
	(0.0112741)
Interstate Mergers	-0.052***
	(0.003)
Panel B: Second stage	Evictions
Branch density	-0.673***
	(0.121)
Sample	Urban
Period	2000-2018
F-test joint sign. (2nd stage)	217.40
F-test instruments (1st stage)	127.71
Hansen test (1st stage)	1.102
Number of tracts	27,926
Observations	288,053

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$. The main explanatory variable is number of branches per inhabitant. The results in Panel A corresponds to the first stage of IV regressions where the instruments for the variable Branches per Inhabitant are the variables Intercounty Mergers and Interstate Mergers. Results in Panel B corresponds to the second stage of IV regressions. All regressions include tract and year fixed effects.

8.4 Alternative measures of evictions

We complete our analysis considering alternative variables that measure the relevance of evictions. The dependent variable in Table 3 is the number of *Households Threatened with Eviction*, which reflects the number of households in the census tract-year that received at least one eviction filling. Note that this variable only accounts for the first (or only) time a household was filled with an eviction in the tract-year. However, Eviction Lab (EL) collects two additional variables that complements this information. Column 2 in Table 9 show the results when considering the variable *Filings*, which sums all evictions filings at the census tract-year level. Moreover, column 3 shows the results when using the variable *Judgement*, which measures the number of eviction filings that resulted in eviction judgments. The results are consistent with those of our baseline model (column 1 in Table 9). We find that a 1% increase in the *Distance* to bank branches in urban tracts increased the total number of eviction filings by 3.2% and the number of eviction judgements by 2.4%.

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Table 9: Effect of distance on evictions: Alternative measures of evictions

	Distance		
Panel A: First stage	(1)	(2)	(3)
Intercounty Mergers	0.019***	0.019***	0.019***
	(0.005)	(0.005)	(0.005)
Interstate Mergers	0.021***	0.021***	0.022***
	(0.010)	(0.001)	(0.001)
Panel B: Second stage	Evictions	Filings	Judgements
Distance	3.005***	3.228***	2.461***
	(0.362)	(0.390)	(0.313)
Sample	Urban	Urban	Urban
Period	2000-2018	2000-2018	2000-2018
F-test joint sign. (2nd stage)	300.72***	293.65***	237.48***
F-test instruments (1st stage)	73.97***	73.96***	73.06***
Hansen test (1st stage)	1.697	1.204	1.599
Number of tracts	44,984	44,984	44,042
Observations	612,958	612,958	563,067

Notes: Robust standard errors, in parentheses, are clustered at the tract level. *** $p < 0.01$, ** $p < 0.05$, *$p < 0.1$. In column (1), the dependent variable is the number of evictions, while columns (2) and (3) considers the number of eviction filings and judgements. In all regressions, the main explanatory variable is Distance. Panel A shows the results of the first stage of the IV regressions, where the instruments for distance are the Intercounty Mergers and Interstate Mergers. Panel B presents the results of the second stage of IV regressions. All regressions include tract and year fixed effects.

9. Conclusions

This paper has examined the impact of financial exclusion on evictions, focusing on how changes in the number of bank branches in the US from 2000 to 2018 affected the number of households threatened with eviction. To overcome potential endogeneity associated to branch closures, we have adopted an instrumental variable (IV) identification strategy, using bank mergers as an instrument of the distance between the population and the closest bank branch at the census tract level. The results of the first stage of the IV estimation show that both interstate and intercounty mergers had a positive and significant effect on the distance to the bank branches. These results imply that, during the period studied, mergers increased financial exclusion by forcing consumers to travel farther for banking services. Similar results are obtained when we use the number of branches per inhabitant as a measure of financial exclusion, with tracts affected by mergers experiencing a reduction of branch density.

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The main result of our analysis is that an increase in the distance to the closest bank branch has a positive and significant impact on the number of households threated with eviction. Specifically, we show that a 1% increase in the distance to the closest branch generated a 2.4% increase in the number of households filled with an eviction. This effect is primarily driven by urban tracts, as we do not find a statistically significant relationship between distance to branches and eviction fillings in rural tracts. These results are robust when we apply a matching procedure to control for pre-existing observable differences between tracts exposed and unexposed to mergers. They also hold when we focus on mergers where the acquirer and the acquired banks had overlapping branches, and when considering alternative measures of evictions, such as the total number of evictions at the census tract-year level and the number of eviction filings resulting in eviction judgments.

Another contribution of this paper is to examine the evolution of the relationship between financial exclusion and evictions over time. We have divided our sample in three sub-periods: 2000-2007 and 2008-2013 and 2014-2018. The first interval corresponds to a period of expansion in bank branches and the other two correspond to a period of branch consolidation after the 2008 financial crisis. Our results show that distance to bank branches had a positive impact on evictions in the 2000-2007 and 2008-2014 periods. Specifically, a 1% increase in the distance to the closest branch generated a 5% and 1.75%, respectively, increase in the number of households threatened with eviction. This suggest that during the branch expansion period (2000-2007), the presence of more branches could increase the financial support received by tenants, reducing the likelihood of evictions. Likewise, during the post-crisis period (2018-2013), branch closing could exacerbate tenants' financial difficulties and make them more exposed to evictions. Finally, we do not find a significant relationship between distance and evictions in the 2014-18 period, although this result may be affected by the limitations of the IV procedure for this subperiod.

Our research shows that financial exclusion is a key driver of tenant's exposition to evictions. The availability and geographic proximity to bank branches shapes credit allocation to tenants and determines their ability to satisfy rents. In this context, bank consolidation and branch closures might have important implications, particularly in cases where credit allocation heavily depends on soft information about borrowers. When a branch closes, the

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loss of soft information about tenants can negatively impact their financial stability and limit their access to adequate financial advice. According to Nguyen (2019) "*distance matters because it facilitates the forging of branch-specific relationships that, once disrupted, make it difficult for borrowers to seamlessly switch to another lender"*. Branch closures affect local credit supply and might leave the urban poor very close to eviction.

Acknowledgements: We are grateful to the participants at the 48th Simposio de la Asociación Española de Economía (Salamanca, 2023) for their valuable comments. We also gratefully acknowledge financial support from the Spanish Ministerio de Ciencia e Innovación (MCI), Agencia Estatal de Investigación (AEI) and the Fondo Europeo de Desarrollo Regional (FEDER) (Grant: PID2021- 128237OB-I00). The usual disclaimer applies.

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